QUBO Formulation: FPGA Placement Problem

FPGA Placement Problem

Problem Statement. Given a set of N logic blocks $\mathcal{B} = \{0, 1, \ldots, N-1\}$ and a set of M discrete FPGA sites $\mathcal{S} = \{0, 1, \ldots, M-1\}$, the goal is to assign each block to a unique site such that the total weighted communication distance between connected blocks is minimized. Each site can accommodate at most one block. A set of connections $\mathcal{C} \subseteq \mathcal{B} \times \mathcal{B}$ defines pairs of blocks that communicate, with associated weights $w_{i,k}$ representing the importance or frequency of communication between block i and block k. The distance between any two sites $j, l \in \mathcal{S}$ is given by a metric $d_{j,l} \geq 0$. The objective is to find a placement that minimizes the sum of weighted distances between all connected block pairs based on their assigned sites.

Decision Variables. We introduce binary variables $x_{i,j} \in \{0, 1\}$ for each block $i \in \mathcal{B}$ and site $j \in \mathcal{S}$. The variable $x_{i,j}$ takes the value 1 if and only if block i is assigned to site j, and 0 otherwise. The total number of variables is $n = N \times M$.

QUBO Derivation. We construct a Quadratic Unconstrained Binary Optimization (QUBO) Hamiltonian $H(\mathbf{x})$ by encoding the objective function and constraints into a single energy function. The QUBO form is $H(\mathbf{x}) = \sum_{u \leq v} Q_{u,v} x_u x_v + c$, where x_u denotes the variable indexed by $u = iM + j$.

Step 1: Objective Function. The objective is to minimize the total weighted wirelength:

$$E_{\text{obj}} = \sum_{(i,k) \in \mathcal{C}} \sum_{j \in \mathcal{S}} \sum_{l \in \mathcal{S}} w_{i,k}\, d_{j,l}\, x_{i,j}\, x_{k,l}. \tag{1}$$

Since QUBO minimizes energy, we set $H_{\text{obj}} = E_{\text{obj}}$.

Step 2: Constraints. The problem imposes two constraints:

1. *Block Assignment:* Each block must be assigned to exactly one site:

$$\sum_{j \in \mathcal{S}} x_{i,j} = 1, \quad \forall i \in \mathcal{B}. \tag{2}$$

2. *Site Capacity:* Each site can hold at most one block:

$$\sum_{i \in \mathcal{B}} x_{i,j} \leq 1, \quad \forall j \in \mathcal{S}. \tag{3}$$

Step 3: Penalty Terms. We convert constraints to penalties using a weight $\lambda > 0$.

Block Assignment Penalty: We square the constraint violation:

$$P_{\text{block}} = \lambda \sum_{i \in \mathcal{B}} \left(\sum_{j \in \mathcal{S}} x_{i,j} - 1 \right)^2 \tag{4}$$

$$= \lambda \sum_{i \in \mathcal{B}} \left(\sum_{j \in \mathcal{S}} x_{i,j}^2 + \sum_{j \neq l \in \mathcal{S}} x_{i,j} x_{i,l} - 2 \sum_{j \in \mathcal{S}} x_{i,j} + 1 \right). \tag{5}$$

Using the idempotent property $x_{i,j}^2 = x_{i,j}$ and noting that $\sum_{j \neq l} x_{i,j} x_{i,l} = 2 \sum_{j < l \in \mathcal{S}} x_{i,j} x_{i,l}$, we expand:

$$P_{\text{block}} = \lambda \sum_{i \in \mathcal{B}} \left(\sum_{j \in \mathcal{S}} x_{i,j} + 2 \sum_{j < l \in \mathcal{S}} x_{i,j} x_{i,l} - 2 \sum_{j \in \mathcal{S}} x_{i,j} + 1 \right) \tag{6}$$

$$= \lambda \sum_{i \in \mathcal{B}} \left(1 - \sum_{j \in \mathcal{S}} x_{i,j} + 2 \sum_{j < l \in \mathcal{S}} x_{i,j} x_{i,l} \right). \tag{7}$$

This yields linear terms $-\lambda x_{i,j}$, quadratic terms $2\lambda x_{i,j} x_{i,l}$ for $j \neq l$, and a constant offset contribution λ.

Site Capacity Penalty: The inequality $\sum_i x_{i,j} \leq 1$ is enforced by penalizing pairs of blocks assigned to the same site:

$$P_{\text{site}} = \lambda \sum_{j \in \mathcal{S}} \sum_{i < i' \in \mathcal{B}} x_{i,j} x_{i',j}. \tag{8}$$

This term is zero if at most one block is assigned to site j, and positive otherwise.

Step 4: Combined Hamiltonian. The total QUBO Hamiltonian is:

$$H(\mathbf{x}) = H_{\text{obj}} + P_{\text{block}} + P_{\text{site}}. \tag{9}$$

Step 5: Q Matrix Entries and Offset. Reading off the coefficients from the expanded form, the QUBO parameters are:

$$Q_{(i,j),(k,l)} = w_{i,k}\, d_{j,l}, \qquad \forall (i,k) \in \mathcal{C},\ j,l \in \mathcal{S}, \tag{10}$$

$$Q_{(i,j),(i,j)} = -\lambda, \qquad \forall i \in \mathcal{B},\ j \in \mathcal{S}, \tag{11}$$

$$Q_{(i,j),(i,l)} = 2\lambda, \qquad \forall i \in \mathcal{B},\ j,l \in \mathcal{S},\ j \neq l, \tag{12}$$

$$Q_{(i,j),(i',j)} = \lambda, \qquad \forall i,i' \in \mathcal{B},\ j \in \mathcal{S},\ i \neq i', \tag{13}$$

$$c = \lambda N. \tag{14}$$

All other entries $Q_{u,v}$ are zero.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any constraint violation increases the energy more than any possible reduction

in the objective function. A sufficient condition is:

$$\lambda > \sum_{(i,k)\in\mathcal{C}} |w_{i,k}|. \tag{15}$$

This guarantees that the minimum penalty cost for violating a constraint exceeds the maximum possible gain achievable by reducing the objective, thereby ensuring that the global minimum of $H(\mathbf{x})$ corresponds to a feasible solution.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^n$ recovers the optimal placement because: (i) for any feasible assignment, the penalty terms vanish, so $H(\mathbf{x}) = E_{\text{obj}}(\mathbf{x})$; (ii) for any infeasible assignment, the penalty terms are strictly positive. Given $\lambda > \sum |w_{i,k}|$, the energy of any infeasible solution exceeds that of the best feasible solution, as the penalty increase dominates any potential decrease in E_{obj}. Thus, the global minimum of $H(\mathbf{x})$ must be feasible and minimize E_{obj}, yielding the optimal placement.

Illustrative Example. Consider a small instance with $N = 2$ blocks $\mathcal{B} = \{0,1\}$ and $M = 2$ sites $\mathcal{S} = \{0,1\}$. There is a single connection $(0,1)$ with weight $w_{0,1} = 1$. The distance metric is $d_{0,1} = d_{1,0} = 1$ and $d_{0,0} = d_{1,1} = 0$. We set $\lambda = 10$.

The variables are $x_{0,0}, x_{0,1}, x_{1,0}, x_{1,1}$ mapped to indices $0, 1, 2, 3$. The objective terms are $x_{0,0}x_{1,1}$ and $x_{0,1}x_{1,0}$, each with coefficient 1.

The block assignment penalty expands to:

$$\begin{aligned} P_{\text{block}} &= 10\left[(x_{0,0} + x_{0,1} - 1)^2 + (x_{1,0} + x_{1,1} - 1)^2\right] && (16)\\ &= 20 - 10(x_{0,0} + x_{0,1} + x_{1,0} + x_{1,1}) + 20(x_{0,0}x_{0,1} + x_{1,0}x_{1,1}). && (17) \end{aligned}$$

The site capacity penalty is:

$$P_{\text{site}} = 10(x_{0,0}x_{1,0} + x_{0,1}x_{1,1}). \tag{18}$$

The total Hamiltonian is:

$$\begin{aligned} H(\mathbf{x}) &= x_{0,0}x_{1,1} + x_{0,1}x_{1,0} + 20 - 10(x_{0,0} + x_{0,1} + x_{1,0} + x_{1,1}) && (19)\\ &\quad + 20(x_{0,0}x_{0,1} + x_{1,0}x_{1,1}) + 10(x_{0,0}x_{1,0} + x_{0,1}x_{1,1}). && (20) \end{aligned}$$

The constant offset is $c = 20$. The optimal feasible solutions are $x_{0,0} = x_{1,1} = 1$ (blocks at sites $0, 1$) and $x_{0,1} = x_{1,0} = 1$ (blocks at sites $1, 0$), both yielding energy $H = 1$. Any infeasible assignment incurs a penalty of at least 10, resulting in energy ≥ 10.